# Kinetic Theory

**Exam code:** 0620 & 0971

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## Solids, liquids & gases

- The properties of solids, liquids, and gases can be explained by the **kinetic theory of matter**
- This model describes matter as consisting of tiny particles that are in constant motion
- The arrangement and movement of these particles are different in each state of matter
  - This leads to their different physical properties

Help me understand

### Solids

- Solids have a **fixed** volume and shape and they have a high density
- The particles are packed very closely together in a fixed and regular pattern
- The particles **vibrate** in position but can't change location

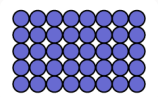
### Liquids

- Liquids also have a fixed volume but adopt the shape of the container
- They are generally less dense than solids (an exception is water), but much denser than gases
- The particles are still close together but are arranged randomly
- The particles **move** and **slide** past each other which is why liquids adopt the shape of the container and are able to flow freely

**Stuck on something?**
Ask questions and get help with exactly what you're revising.
Got it

the shape of the container
- Gases have a very low density
- The particles are far apart and move randomly and quickly (around 500 m/s) in all directions.
- Since there is a lot of space between the particles, gases can be easily compressed.
- They **collide** with each other and with the sides of the container (this is how **pressure** is created inside a can of gas)

## Summary of the properties of solids, liquids and gases

| | Solid | Liquid | Gas |
|---|---|---|---|
| **Diagram** |  | | |
| **Arrangement of particles** | Regular arrangement | Randomly arranged | Randomly arranged |
| **Movement of particles** | Vibrate about a fixed position | Move around each other | Move quickly in all directions |
| **Closeness of particles** | Very close | Close | Far apart |
| **Density** | High | Medium | Low |
| **Energy of particles** | Low energy | Greater energy | Highest energy |

### Examiner Tips and Tricks

In your exam, you will need to apply the **kinetic theory of matter** to explain the properties of solids, liquids, and gases.

This means focusing on the key differences in particle **arrangement**, **movement**, and the **energy** of the particles in each state.

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